Exam 2:-

C6:

(A) $\int x^2 \cos(3x)\,dx$

$u = x^2$	$dv = \cos(3x)\,dx$
$du = 2x\,dx$	$v = \frac{\sin(3x)}{3}$

$\int u\,dv = uv - \int v\,du$

$= \frac{x^2 \sin(3x)}{3} - \int \frac{\sin(3x)}{3}(2x)\,dx + C$

$= \frac{x^2 \sin(3x)}{3} - \frac{2}{3}\int x \sin(3x)\,dx + C$

$= \frac{x^2 \sin(3x)}{3} - \left(\frac{2}{3}\right)\int x \sin(3x)\,dx + C$

$u = x$	$dv = \sin(3x)\,dx$
$du = dx$	$v = -\frac{\cos(3x)}{3}$

$= \frac{x^2 \sin(3x)}{3} - \left(\frac{2}{3}\right)\left[\frac{-x\cos(3x)}{3} - \int \frac{-\cos(3x)}{3}\,dx\right]$

$= \frac{x^2 \sin(3x)}{3} + \left(\frac{2}{3}\right)\frac{x\cos(3x)}{3} - \left(\frac{2}{3}\right)\left(\frac{1}{3}\right)\left(\frac{\sin(3x)}{3}\right) + C$

$$\int x^2 \cos(3x)\,dx = \frac{x^2 \sin(3x)}{3} + \frac{2x\cos(3x)}{9} - \left(\frac{2}{27}\right)\sin(3x) + C$$

(B) $\int x^4 \ln(x)\,dx$

$u = \ln(x)$ | $dv = x^4\,dx$

$du = \frac{1}{x}\,dx$ | $v = \frac{x^5}{5} + C$

$\int u\,dv = uv - \int v\,du$

$= \frac{x^5}{5}\ln(x) - \int \frac{x^5}{5}\left(\frac{1}{x}\right)dx$

$= \frac{1}{5}\left[x^5 \ln(x) - \int x^4\,dx\right]$

$$\int x^4 \ln(x)\,dx = \frac{1}{5}\left[x^5 \ln(x) - \frac{x^5}{5}\right] + C$$

C7.

(A) $\int \sin^3 x \cos^2 x\,dx = \int \cos^2 x \sin^2 x \sin x\,dx$

$= \int (1 - \cos^2 x)\cos^2 x \sin x\,dx$

Let $u = \cos x$, $du = -\sin x\,dx$

$\Rightarrow -du = \sin x\,dx$

$$= \int (1-u^2)\, u^2\, (-du)$$

$$= -\int (u^2 - u^4)\, du$$

$$= -\frac{u^3}{3} + \frac{u^5}{5} + C$$

$$= -\frac{\cos^3 x}{3} + \frac{\cos^5 x}{5} + C$$

$$\therefore \int \sin^3 x \cdot \cos^2 x\, dx = -\frac{\cos^3 x}{3} + \frac{\cos^5 x}{5} + C$$

(B) $\int \tan^4 x\, dx$

$$= \int \tan^2 x \tan^2 x\, dx$$

$$= \int \tan^2 x\, (\sec^2 x - 1)\, dx$$

Let $u = \tan x$, $du = \sec^2 x\, dx$

$$= \int \tan^2 x \sec^2 x\, dx - \int (\sec^2 x - 1)\, dx$$

$$= \int u^2\, du - \int \sec^2 x\, dx + \int dx + C$$

$$= \frac{u^3}{3} - \tan x + x + C$$

$$= \frac{\tan^3 x}{3} - \tan x + x + C$$

$$\therefore \int \tan^4 x \, dx = \frac{\tan^3 x}{3} - \tan x + x + C$$

C:8

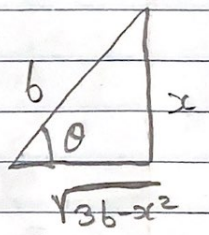
(A)

(i) $\int \sqrt{36 - x^2} \, dx$

$x = 6\sin\theta$

$\sin\theta = x/6$

$dx = 6\cos\theta \, d\theta$



$$= \int \sqrt{36 - 36\sin^2\theta} \; (6\cos\theta \, d\theta)$$

$$= \int 36\cos\theta(\cos\theta) \, d\theta$$

$$= \int 36\cos^2\theta \, d\theta$$

Let $\theta = \sin^{-1}(x/6)$

$d\theta = \left(\frac{dx}{6}\right)$

$$= \int 36\,(1 - \sin^2\theta) \, d\theta$$

$$= 36\left[\int 1 \, d\theta - \int \sin^2\theta \, d\theta\right]$$

$$= 36\left[\theta - \int \left(\frac{1 + \sin(2\theta)}{2}\right) d\theta\right]$$

$$= 36\theta - 18\int 1 \, d\theta + 18\int \sin 2\theta \, d\theta$$

$$= 18\theta - 18\left(\frac{-\cos 2\theta}{2}\right) + C$$

$$= 18\theta + 9\cos 2\theta + C$$

$$= 18\theta + 9\left(\frac{2}{1}\right)(\sin\theta\cos\theta) + C$$

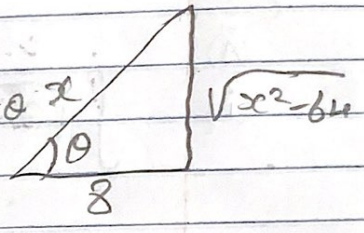
$$= 18\sin^{-1}\left(\frac{x}{9}\right) + 18\left(\frac{x}{6}\right)\left(\frac{\sqrt{36-x^2}}{6}\right) + C$$

$$= 18\sin^{-1}\left(\frac{x}{9}\right) + \left(\frac{1}{2}\right)(x)(\sqrt{36-x^2}) + C$$

(ii) $\int \frac{dx}{x\sqrt{x^2-64}}$

$x = 8\sec\theta$

$dx = 8\sec\theta\tan\theta\, d\theta$



$$= \int \frac{8\sec\theta\tan\theta\, d\theta}{8\sec\theta\,(\sqrt{64\sec^2\theta - 64})}$$

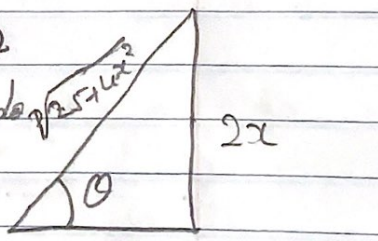
$$= \int \frac{\tan\theta\, d\theta}{8\tan\theta} = \int \frac{1}{8}\, d\theta$$

$$= \frac{1}{8}\theta = \frac{1}{8}\sec^{-1}\left(\frac{x}{8}\right)$$

(iii) $\int \frac{x}{(25+4x^2)^{3/2}}\, dx$

$2x = 5\tan\theta$

$dx = \frac{5}{2}\sec^2\theta\, d\theta$



$$= \int \frac{\left(\frac{5}{2}\right)\tan\theta\left(\frac{5}{2}\right)\sec^2\theta\, d\theta}{(25+25\tan^2\theta)^{3/2}}$$

$$= \int \left(\frac{25}{4}\right)\left(\frac{\tan\theta \sec^2\theta \, d\theta}{125}\right)\left(\frac{1}{\sec^3\theta}\right)$$

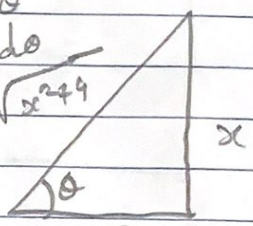
$$= \left(\frac{1}{20}\right)\int \frac{\tan\theta \, d\theta}{\sec^3\theta}$$

(B)

$$\int \frac{dx}{(x^2+9)^{3/2}}$$

$x = 3\tan\theta$

$dx = 3\sec^2\theta \, d\theta$



$$= \int \frac{3\sec^2\theta \, d\theta}{(9\tan^2\theta + 9)^{3/2}}$$

$$= \int \left(\frac{3}{27}\right) \cdot \frac{\sec^2\theta \, d\theta}{\sec^3\theta}$$

$$= \frac{1}{9}\int \frac{1}{\sec\theta} \, d\theta = \frac{1}{9}\int \cos\theta \, d\theta$$

$$= \left(\frac{1}{9}\right) \sin\theta + C$$

$$= \left(\frac{1}{9}\right)\left(\frac{x}{\sqrt{x^2+9}}\right) + C$$

$$\int \frac{dx}{(x^2+9)^{3/2}} = \left(\frac{1}{9}\right)\left(\frac{x}{\sqrt{x^2+9}}\right)$$

C9:

(A) $f(x) = \dfrac{x+10}{(x^2-1)(x^2+9)}$

$$\frac{x+10}{(x^2-1)(x^2+9)} = \frac{x+10}{(x-1)(x+1)(x^2+9)}$$

$$= \frac{A}{(x-1)} + \frac{B}{(x+1)} + \frac{Cx+D}{(x^2+9)}$$

(B) $$\int \frac{x^2+2x-1}{x^3-x}\,dx = \int \frac{x^2+2x-1}{x(x^2-1)}\,dx$$

$$= \int \frac{x^2+2x-1}{x(x-1)(x+1)}\,dx$$

$$x^2+2x-1 = A(x-1)(x+1) + B(x(x+1)) + C(x(x+1))$$

$$x^2+2x-1 = Ax^2 - A + Bx^2 + Bx + Cx^2 - Cx$$

$\therefore$ $A+B+C = 1$

$B-C = 2$

$-A = -1$

$\boxed{A = 1}$

$1 - C = 2$

$\boxed{C = -1}$

$\Rightarrow$ $B+C = 0$

$B-C = 2$

$2B = 2$

$\boxed{B = 1}$

$\boxed{C = -1}$

$$= \int \left[\frac{1}{x} + \frac{1}{(x-1)} - \frac{1}{(x+1)} \right] dx$$

$$= \ln|x| + \ln|(x-1)| - \ln|x+1| + C$$

C10:

(A) $$\int \frac{\sin 2\theta}{\sqrt{5-\sin\theta}}\, d\theta = \int \frac{2\sin\theta\cos\theta}{\sqrt{5-\sin\theta}}\, d\theta$$

Let $u = \sin\theta$, $du = \cos\theta\, d\theta$

$$= \int \frac{2u\, du}{\sqrt{5-u}}$$

Let $a = 5-u$
$da = -du$
$-u = a-5$
$u = 5-a$

$$= \int \frac{2(5-a)(-da)}{\sqrt{a}}$$

$$= -2\int \frac{5}{\sqrt{a}}\, da + 2\int \frac{a}{\sqrt{a}}\, da$$

$$= -10\int (a)^{-1/2}\, da + 2\int \sqrt{a}\, da$$

$$= -10(2\sqrt{a}) + \left(\frac{4}{3}\right)(a^{3/2}) + C$$

$$= -20\left(\sqrt{5-u}\right) + \left(\frac{4}{3}\right)(5-u)^{3/2} + C$$

$$= (20)\sqrt{5 - \sin\theta} + \left(\frac{4}{3}\right)(5 - \sin\theta)^{3/2} + C$$

(B) $\int \sqrt{x}\, e^{\sqrt{x}}\, dx$

Let $u = \sqrt{x}$, $du = \frac{1}{2\sqrt{x}}\, dx$

$\Rightarrow 2u\, du = dx$

$$\int u e^{u}\, 2u\, du = \int 2u^2 e^{u}\, du$$

$$= 2\int u^2 e^{u}\, du$$

A	dB
u^2	e^u
$2u$	e^u
2	e^u
0	e^u

$$= 2\left[u^2 e^{u} - 2ue^{u} + 2e^{u}\right] + C$$

$$= 2\left[xe^{\sqrt{x}} - 2\sqrt{x}\, e^{\sqrt{x}} + 2e^{\sqrt{x}}\right] + C$$

$$= 2xe^{\sqrt{x}} - 4\sqrt{x}\, e^{\sqrt{x}} + 4e^{\sqrt{x}} + C$$

(11) $$\int_2^{\infty} e^{-3x}\,dx = \int_2^{\infty} \frac{1}{e^{3x}}\,dx$$

$$= \lim_{b\to\infty} \int_2^{b} e^{-3x}\,dx$$

$$= \lim_{b\to\infty} \left[\frac{e^{-3x}}{-3}\right]_2^{b}$$

$$= \lim_{b\to\infty} \frac{e^{-3b}}{-3} + \frac{e^{-3(2)}}{3}$$

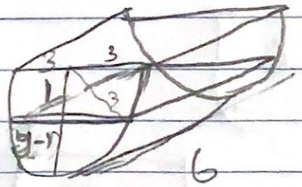
$$= \lim_{b\to\infty} \frac{e^{-6}}{3} - \frac{e^{-3b}}{3}$$

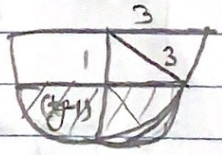
$$= \frac{e^{-6}}{3} - 0$$

$$= \frac{e^{-6}}{3} = \frac{1}{3e^{6}}$$

∴ it is converges

(5:-



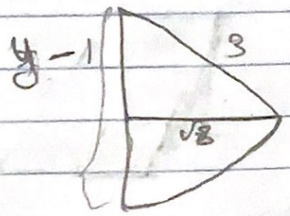


$$W = \int_0^{6} \rho g\,(\text{Area of cross section})\,(\text{Distance out the tank})\,dy$$

$$W = \int_0^6 (1000)(9.8)(\text{Area})(\text{Distance})\, dy$$

Area of Semicircle (Crossection) $= \frac{1}{2}\pi r^2$ | $r = 4$

$$= \frac{1}{2}\pi 8$$



$$W = \int_0^6 (1000)(9.8)(8\pi)(y-1)\, dy$$

$$W = 78400\pi \int_0^6 (y-1)\, dy$$

$$W = 78400\pi \left[\frac{y^2}{2} - y\right]_0^6$$

$$W = 78400\pi \left[\frac{36}{2} - 6\right]$$

$$\boxed{W = 940800\pi \text{ J}} \quad \text{or} \quad \boxed{W = 2955610.368 \text{ J}}$$